

ALGEBRAIC STRUCTURE:
Coset, Normal Subgroup, Group
Homomorphism

Lecture—5-6

ALGEBRAIC STRUCTURE



Coset

➤ let H be a subgroup of a group G , then left coset $aH = \{a * h : a \in G \text{ and } h \in H\}$

Similarly right coset $Ha = \{h * a : a \in G \text{ and } h \in H\}$

➤ Example: if binary operation is addition then

$$H + a = \{h + a : h \in H\}$$

$$a + H = \{a + h : h \in H\}$$

$$\underline{\mathbb{Z}} = \{ \dots, -3, -2, -1, 0, 1, 2, 3, \dots \}$$

$$H = \{ \dots, -9, -6, -3, 0, 3, 6, 9, \dots \}$$

$$\rightarrow H+0 = \{ \dots, -9, -6, -3, 0, 3, 6, 9, \dots \}$$

$$\rightarrow H+1 = \{ \dots, -8, -5, -2, 1, 4, 7, 10, \dots \}$$

$$\rightarrow H+2 = \{ \dots, -7, -4, -1, 2, 5, 8, 11, \dots \}$$

$$\rightarrow \underline{\underline{H+3}} = \{ \dots, -6, -3, 0, 3, 6, 9, \dots \}$$

Coset

Example: let G be additive group of integers ,that is $G = \{ \dots -3, -2, -1, 0, 1, 2, 3 \dots \}$

Let H be the subgroup of obtained G on multiplying each element of G by 3. then

$$H = \{ \dots -9, -6, -3, 0, 3, 6, 9 \dots \}$$

Then right coset of H in G are

$$H+0 = \{ -9, -6, -3, 0, 3, 6, 9 \dots \}$$

$$H+1 = \{ -8, -5, -2, 1, 4, 7, 10 \dots \}$$

$$H+2 = \{ -7, -4, -1, 2, 5, 8, 11, \dots \}$$

We see the right cosets $\{H + 0, H + 1, H + 2\}$ are all distinct and disjoint

$$G = \{ \dots -3, -2, -1, 0, 1, 2, 3, 4, \dots \} +$$

$$H = \{ \dots -9, -6, -3, 0, 3, 6, 9, \dots \} +$$

$$\underline{L \oplus H \text{ coset}}$$

$$a + H, a \in G, H \in H$$

$$0 + H = \{ \dots -9, -6, -3, 0, 3, 6, 9, \dots \}$$

$$1 + H = \{ \dots -8, -5, -2, 1, 4, 7, 10, \dots \}$$

$$\mathbb{Z} = \{\dots, -4, -3, -2, -1, 0, 1, 2, 3, 4, \dots\}$$

$$H = \{\dots, -9, -6, -3, 0, 3, 6, 9, 12, \dots\}$$

$$\begin{array}{l} H+0 \\ \hline \end{array} = \{\dots, -9, -6, -3, 0, 3, 6, 9, 12, \dots\}$$

$$\begin{array}{l} H+1 \\ \hline \end{array} = \{\dots, -8, -5, -2, 1, 4, 7, 10, \dots\}$$

$$\begin{array}{l} H+2 \\ \hline \end{array} = \{\dots, -7, -4, -1, 2, 5, 8, 11, 3, \dots\}$$

$$H+3 = \times$$

$$H+4 \quad \text{left coset} = \text{right coset} //$$

Coset

What about $H+3, H+4, H+5, \dots$

Note: In abelian group each left coset equal to right coset.

Properties of cosets: (Home work) Proof the following

Let H be a subgroup of G , and a, b belong to G then,

1. $a \in aH$
2. $aH = H$ if and only if $a \in H$
3. $aH = bH$ or $aH \cap bH = \emptyset$

Or

Let aH and bH be two cosets of H . either aH and bH are disjoint or they are identical.

4. $aH = bH$ if and only if $a^{-1}b \in H$

Lagrange's Theorem

- If G is finite group and H is a subgroup of G , then

$$|H| \text{ divides } |G|.$$

$\equiv$

*$|H|$: order of H
 $|G|$: order of G .*

The number of distinct left (right) cosets of H in G is $|G|/|H|$

$$\begin{aligned} G &= \{1, -1, 2, -2\} \\ H &= \{1, -1\} \end{aligned}$$

✓

Proof:

- Let $H \leq G$ with $|G| = n$, and $|H| = k$.
- Write the elements of H in row 1:

e	h_2	h_3	$\dots$	h_k
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Proof:

- Choose any a_2 in G not in row 1.
- Write a_2H in the second row.

e	h_2	h_3	$\dots$	h_k
a_2	$a_2 h_2$	$a_2 h_3$	$\dots$	$a_2 h_k$

Proof:

- Continue in a similar manner...
- Since G is finite, this process will end

e	h_2	h_3	$\dots$	h_k
a_2	$a_2 h_2$	$a_2 h_3$	$\dots$	$a_2 h_k$
a_3	$a_3 h_2$	$a_3 h_3$	$\dots$	$a_3 h_k$
$\vdots$	$\vdots$	$\vdots$		$\vdots$
a_r	$a_r h_2$	$a_r h_3$		$a_r h_k$

$$O(n) = \leftarrow \Rightarrow O(n)/O(1)$$

Proof:

- Rows are disjoint by
- Each row has k elements by

e	h_2	h_3	...	h_k
a_2	$a_2 h_2$	$a_2 h_3$	...	$a_2 h_k$
a_3	$a_3 h_2$	$a_3 h_3$	...	$a_3 h_k$
$\vdots$	$\vdots$	$\vdots$		$\vdots$
a_r	$a_r h_2$	$a_r h_3$		$a_r h_k$

$$O(n) =$$

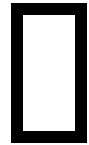
$$= \underline{k}$$

$$O(1) = \underline{k}$$

$$\Rightarrow O(n) / O(1) = \frac{n}{k} = \underline{\underline{k}}$$

My proof:

- Let r be the number of distinct cosets.
- Clearly $|G| = |H| \cdot r$, and $r = |G|/|H|$.



e	h_2	h_3	$\dots$	h_k
a_2	$a_2 h_2$	$a_2 h_3$	$\dots$	$a_2 h_k$
a_3	$a_3 h_2$	$a_3 h_3$	$\dots$	$a_3 h_k$
$\vdots$	$\vdots$	$\vdots$		$\vdots$
a_r	$a_r h_2$	$a_r h_3$		$a_r h_k$

Normal subgroups

- **Definition 18:** A subgroup H of a group is a normal subgroup if $gH=Hg$ for $\forall g \in G$.
- **Example:** Any subgroups of Abelian group are normal subgroups.

Normal subgroups

- Theorem: A subgroup H of a group G is normal if and only if $g^{-1}hg \in H$ for every $h \in H, g \in G$.

- Proof: Let H be a normal subgroup of a group G for every $h \in H, g \in G$.

Then $Hg = gH$ (by definition)

Now $hg \in Hg = gH$

Normal subgroups

SO $hg = gh_1$ for some $h_1 \in H$

That is $\underline{g^{-1}hg} = h_1 \in H$

Conversely $g^{-1}hg \in H$ for every $h \in H$ we have to show that $\underline{Ha = aH}$

i.e $\underline{Ha \subseteq aH}$ and $\underline{aH \subseteq Ha}$

$$\begin{aligned} & Ha \subseteq aH \\ & aH \subseteq Ha \end{aligned}$$

Normal subgroups

Consider $a \in G$ for any $h \in H$, $a^{-1}ha \in H$
 $\therefore ha = a(a^{-1}ha) \in aH = ah_1 \in aH$
 $\therefore Ha \subseteq aH$

Similarly

$$ah = (aha^{-1})a \in Ha \Rightarrow aH \subseteq Ha$$

Hence $aH = Ha$

$$\therefore Ha \subseteq aH$$

Homomorphism of Group

Let $(G, .)$ and $(H, *)$ be two groups and f be function from G into H . Then f is called a homomorphism of G into H if for all $a, b \in G$

$$f(a.b) = f(a) * f(b)$$

$$f(a.b) = f(a) * f(b)$$

A Homomorphism f of a group G into a group G_1 is called a isomorphism of G onto G_1 if f is one-one and onto.

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Homomorphism of Group

Example: Let $G=\mathbb{R}$ be the group of real number under addition and $G^1 = \mathbb{R}^+$, *the group of positive real numbers* for multiplication. The mapping $f: G \rightarrow G^1$ given by $f(a) = 2^a$ is a group homomorphism because

$$f(a + b) = 2^{a+b} = 2^a \cdot 2^b = f(a) \cdot f(b)$$